

CAPSTONE PROJECT REPORT

Zomato Restaurant Analytics

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Project Overview :

This project involves a comprehensive analysis of Zomato's global restaurant data. The primary goal is to perform end-to-end data processing—from raw data cleaning to advanced feature engineering—to derive business insights using different Analytics tool (**Python,SQL,Power BI**) regarding restaurant performance, pricing strategies, and consumer preferences across different countries.

Project Objectives :

- To standardize multi-currency data into a single currency (INR) for global comparison.
- To clean and normalize complex "wide-format" cuisine data into a usable format.
- To identify high-performing restaurant segments based on ratings and costs.
- To create a robust, analysis-ready dataset for an interactive Power BI dashboard.

Data Dictionary (Merged Dataset)

After performing a **Left Join** between the **Restaurant Data** and **Country Lookup** table on **CountryCode**, the following columns form the core of our analysis:

Attribute Name	Data Type	Description
RestaurantID	Integer	A unique numerical identifier assigned to each restaurant in the database.
RestaurantName	String	The official registered name of the restaurant.
CountryCode	Integer	A numerical code representing the country (used as a Join Key).
Country	String	The name of the country where the restaurant is located (e.g., India, USA, UAE).
City	String	The specific city or metropolitan area of the restaurant.
Address	String	The complete physical street address of the establishment.
Locality	String	The specific neighborhood or district within the city.
LocalityVerbose	String	An extended description of the locality, often including the city name.
Longitude	Float	Geographic coordinate representing the East-West position of the restaurant.
Latitude	Float	Geographic coordinate representing the North-South position of the restaurant.
Cuisines	String	A list of food styles served (e.g., Italian, Chinese). Contains 9 null values.
Cuisines 1 to 8	String	Supplementary columns where cuisines were pre-split into individual attributes.
Average_Cost_for_two	Float	The estimated price for a meal for two people in the local currency.
Currency	String	The specific currency used in the restaurant's region (e.g., INR, Dollar, Pounds).
Has_Table_booking	String	A binary flag ("Yes"/"No") indicating if the restaurant accepts reservations.
Has_Online_delivery	String	A binary flag ("Yes"/"No") indicating if the restaurant offers online ordering.
Is_delivering_now	String	Real-time indicator of whether the restaurant is currently accepting delivery orders.
Switch_to_order_menu	String	A feature flag indicating if a direct "Order Menu" toggle is available on the app.
Price_range	Integer	A categorical cost ranking (1 to 4) provided by the original data source.
Aggregate_rating	Float	The weighted average customer rating on a scale of 0 to 5.0.
Votes	Integer	The total number of individual customer reviews or ratings submitted.

Data Preprocessing & Cleaning

In this phase, we move from raw, unrefined data to a "Clean State." The primary objective here is **Data Hygiene**—ensuring the dataset is accurate, consistent, and logically structured for analysis. Without this phase, your statistical operations (like averages or correlations) would be skewed by incorrect formats, data type mismatches, or non-informative noise.

1. Dimensionality Reduction (Noise Removal)

- **Description:** We identified and dropped irrelevant columns such as `Switch_to_order_menu`, `Country Code`, and internal technical IDs like `Country ID`.
- **Logic:** Since these columns contained redundant data or constant values with zero variance, removing them reduced the dataset's complexity and improved overall memory efficiency.

2. Structural Normalization (Wide-to-Long Format)

- **Description:** The cuisine data was scattered across 8 separate columns (`Cuisines 1` to `Cuisines 8`).
- **Logic:** Having multiple sparse columns made it impossible to perform aggregate functions. We merged these into a single, unified `All_Cuisines` column to facilitate easier grouping and analysis of popular food trends.

3. Data Type Standardization & Audit

- **Description:** We conducted a comprehensive schema audit to ensure all features were stored in their appropriate format.
- **Logic:** We converted the `DateKey` column into a `DateTime` format to enable temporal analysis. While columns like `Votes` and `Average_Cost_for_two` were already numeric, we verified their sub-types (e.g., integer vs. float) to ensure computational precision for all future arithmetic operations.

4. Data Quality & Missing Value Management

- **Description:** We scrutinized the dataset for null values, specifically focusing on critical target variables like **Rating**.
- **Logic:** To maintain analytical integrity, we removed records where the **Rating** was missing, as these were statistically invalid. For non-essential categorical missing values, we performed imputation to handle gaps without creating bias.

5. Initial Statistical Profiling

- **Description:** We performed descriptive statistical operations (Mean, Median, Standard Deviation, Min/Max) across all numeric features.
- **Logic:** This acted as a diagnostic check to detect extreme outliers and verify data distribution. This step ensured that our dataset was statistically sound and reliable before proceeding to deeper bivariate analysis or feature engineering.

Exploratory Data Analysis (EDA)

In this phase, we analyzed the raw columns to identify "Noise" and "Inconsistencies." Here is the logical justification for the pruning process:

1. Data Structural Validation

- **Dimensions Assessment:** We utilized the `.shape` attribute to identify the total number of records and features, ensuring the dataset size is sufficient for robust statistical modeling.
- **Data Type Mapping:** We audited the schema to verify that numerical columns (like `Votes`) and categorical columns (like `City`) were correctly identified, which is crucial for choosing the right visualization libraries.
- **Completeness Audit:** We mapped the extent of missing values across the dataset to determine if the gaps were isolated in specific columns or spread across the entire table, informing our final cleaning strategy.

2. Univariate Analysis (Individual Feature Insight)

- **Numerical Distribution:** We generated histograms for `Votes` and `Average_Cost_for_two` to check for skewness; this revealed whether the data is normally distributed or heavily impacted by extreme values.
- **Categorical Cardinality:** We used `nunique()` and `value_counts()` on features like `City` and `Cuisines` to determine which categories are dominant and which are statistically insignificant ("the long tail").
- **Rating Frequency:** We analyzed the frequency distribution of `Rating` scores to define the "Baseline" (or average) restaurant

performance, which helps in identifying what constitutes an "Elite" restaurant later on.

3. Bivariate Analysis (Relationship Mapping)

- **Popularity vs. Quality:** We plotted `Votes` against `Rating` to determine if higher-rated restaurants naturally command more customer engagement or if popularity is driven by other factors.
- **Pricing vs. Satisfaction:** We visualized `Average_Cost_for_two` vs. `Rating` to see if premium pricing is a reliable indicator of quality (high rating), or if budget-friendly restaurants perform just as well.
- **Delivery Impact:** We compared the average rating of restaurants with `Has_Online_Delivery` enabled versus those that do not, testing the hypothesis that digital presence correlates with better service scores.

4. Statistical Anomaly & Trend Detection

- **Skewness Analysis:** We identified that `Votes` followed a Pareto-like distribution (80/20 rule), where a tiny fraction of restaurants captured the majority of user votes, indicating a "Winner-Takes-All" market dynamic.
- **Outlier Detection:** We utilized Boxplots to identify extreme `Average_Cost_for_two` values, separating "Luxury" outliers from the standard market range to ensure our averages remain representative.
- **Categorical Segmentation:** We investigated the relationship between `Price_range` and `Rating` to understand if different price brackets maintain consistent quality, highlighting distinct segments for targeted business strategy.

Feature Engineering

In this phase, we move beyond exploration into Feature Engineering, where we transform raw attributes into powerful, business-centric indicators. By creating synthetic metrics like standardized pricing and elite performance flags, we simplify complex data relationships, allowing for faster, deeper actionable insights in our final reports.

1. Currency Standardization & Normalization

- **Exchange Rate Integration:** We implemented a function to map the `Currency` column against current exchange rates to normalize all `Average_Cost_for_two` values into a uniform `INR` format.
- **Unified Pricing Scale:** By converting international currencies to a single standard, we eliminated local economic bias, enabling a direct and fair comparison of restaurant pricing models across global cities.
- **Economic Comparability:** This creates a standard baseline for price-performance analysis, ensuring that a "luxury" restaurant in one region can be accurately compared to a "luxury" restaurant in another.

2. Rating Interpretation & Categorical Encoding

- **Dynamic Rating Mapping:** We developed a function to map raw numerical `Rating` values into categorical `Rating_Text` (e.g., Excellent, Very Good, Average) and corresponding color codes.
- **Stakeholder Interpretability:** Converting abstract floats (like 4.6) into human-readable descriptors significantly enhances the dashboard's usability for non-technical stakeholders.
- **Categorical Feature Creation:** We transformed continuous rating scores into a categorical feature, allowing for easier filtering and grouping when analyzing "Customer Satisfaction" levels.

3. Balanced Price Segmentation via Quantile Binning

- **Application of `qcut`:** We utilized the `qcut` function to create a new feature, `new_price_range_category`, which partitions the restaurants into equal-sized bins (e.g., Low, Medium, High, Premium).
- **Statistical Logic:** Raw pricing data often suffers from extreme skewness, where most restaurants are clustered in the low-cost range. Using equal-frequency binning ensures that each category has a statistically significant number of samples, preventing skewed visual representations.
- **Analytical Advantage:** This creates balanced segments that allow us to conduct fair "Apples-to-Apples" comparisons, ensuring our analysis of price-to-rating correlations isn't dominated by the sheer volume of budget-friendly restaurants.

4. Synthetic "Elite Cluster" Business Metrics

- **Elite Identification Logic:** We engineered a synthetic `Elite_Cluster` flag by aggregating multiple high-performance conditions (Rating > 4.2 AND Online Delivery Enabled AND Table Booking Enabled).
- **Operational Performance Metric:** This complex feature acts as a "Gold Standard" filter, allowing us to instantly identify high-value market segments without writing complex, redundant multi-condition queries during analysis.
- **Market Opportunity Scoring:** By synthesizing these metrics, we created a new variable that highlights "High-Value Cities," providing an immediate snapshot of which regions offer the highest potential for business expansion.

SQL-Based Analytical Insights

In this phase, we transition from data preparation in Python to SQL-based database analysis, leveraging the power of relational database management systems to derive complex business insights. By migrating our cleaned dataset into MySQL Workbench, we executed a series of advanced queries that not only showcase technical proficiency in SQL but also provide critical answers to real-world business challenges.

1. Data Migration & Integration

- **Description:** Exported the finalized, cleaned dataset from Python and imported it into MySQL Workbench to establish a relational schema.
- **Logic:** While Python is superior for cleaning and preprocessing, SQL is the industry standard for querying and extracting specific business insights at scale. This migration ensures the data is in an environment optimized for complex joins and multi-dimensional analysis.

2. Architecture of Advanced Querying

- **Description:** Executed 35 complex queries utilizing advanced SQL constructs, including CTEs (Common Table Expressions), Window Functions (`DENSE_RANK`, `LEAD`, `LAG`), and Aggregation logic.
- **Logic:** Simple `SELECT` statements are insufficient for deep analysis. By using ranking functions and subqueries, we were able to perform sophisticated tasks like identifying competitive "runner-up" restaurants and segmenting the market based on performance benchmarks.

3. Relational Relationship Modeling

- **Description:** Implemented complex **Self-Joins** and **Multi-Table Joins** to analyze restaurant relationships and market saturation.
- **Logic:** Raw data contains hidden connections. By joining the table against itself, we uncovered patterns like "Nearby Competitors" and "Brand Duplication" (restaurants with same cuisine and rating in the same city), which would be computationally expensive to compute in a non-relational environment.

4. Business Insight Extraction

- **Description:** Translated technical queries into business-centric metrics, such as "Digital Revenue Share," "Market Cannibalization," and "Elite Cluster Identification."
- **Logic:** An analyst's value lies in the "So What?". Each query was designed to answer a specific business question—such as identifying cities ripe for expansion or highlighting "Elite" restaurants that represent Zomato's high-value user base.

Key Business Insights

(Based on SQL Queries)

I executed 35 complex SQL queries—utilizing advanced window functions and CTEs—to extract 35 distinct analytical findings from the cleaned dataset. To prioritize strategic clarity for stakeholders, I performed an analytical synthesis to distill these into the 10 most impactful business insights that drive meaningful growth. This process demonstrates my ability to navigate technical complexity while maintaining a constant focus on high-level business value.

1. Market Dominance & Geographic Hubs

- **Insight:** Identification of high-listing volume countries and cities.
- **Business Value:** Defines Zomato's "Revenue Engines." These regions serve as the primary operational network and should be the focal point for high-ROI marketing campaigns and premium service rollouts.

2. Premium Market Segmentation

- **Insight:** Pinpointing clusters with high average dining costs and luxury outlets.
- **Business Value:** Enables targeted strategies for HNI (High-Net-Worth Individual) segments. These areas are ideal for launching exclusive subscription services like "Zomato Gold" and curating luxury dining collections.

3. Digital Maturity Index

- **Insight:** Cities with high "Online Delivery" penetration rates.
- **Business Value:** Identifies "Digitally Mature" markets where consumer behavior is perfectly aligned with rapid infrastructure scaling. These are the most profitable zones for hyper-local advertising and delivery fleet expansion.

4. High-Conversion "Value Gems" Strategy

- **Insight:** Correlation of high ratings with low average costs.
- **Business Value:** These restaurants are the primary drivers for new user acquisition and repeat orders. We can leverage these as "high-converting marketing collections" to appeal to the largest consumer segment: the budget-conscious, quality-seeking user.

5. Market Expansion Opportunities

- **Insight:** Detection of regions with high total activity but low restaurant density.
- **Business Value:** Highlights "Untapped Regions" where the market demand exceeds supply. This provides a clear roadmap for vendor-onboarding teams to launch targeted acquisition campaigns and increase market share.

6. Brand Advocacy & Influencer Identification

- **Insight:** Restaurants with exceptional vote-to-rating ratios.
- **Business Value:** Identifies "Market Influencers" or restaurants with organic brand loyalty. Zomato can leverage these specific partners for exclusive brand-led marketing campaigns to boost platform engagement.

7. Quality & Reliability Benchmarking

- **Insight:** Restaurants that maintain both high ratings and a massive volume of votes.
- **Business Value:** Establishes a standard for "Verified Excellence." Showcasing these top-tier performers builds critical user trust, driving higher conversion rates for new app users who are hesitant to try new places.

8. Cuisine-Based Demand Forecasting

- **Insight:** Analysis of the most popular cuisine combinations across various demographics.
- **Business Value:** Provides data-driven consultancy for new restaurant partners. By recommending high-demand menu types, we improve vendor success rates and reduce the "failure-to-launch" risk for new partners.

9. New Vendor Lifecycle Support ("Cold Start" Problem)

- **Insight:** Highlighting restaurants with 0.0 or low ratings that are new to the platform.
- **Business Value:** Proactive support for new vendors. By identifying these outlets, Zomato can provide promotional boosts to help them secure their first reviews, effectively solving the "Cold Start" problem and increasing overall platform health.

10. Historical Market Velocity

- **Insight:** Tracking the peak years of restaurant openings and historical growth bursts.
- **Business Value:** Allows the strategy team to correlate growth with external factors (e.g., economic stability or infrastructure). This longitudinal analysis helps predict future expansion cycles and informs long-term capital allocation.

Strategic Recommendations for Zomato

Business Growth

1. Market Dominance: Since these regions serve as your "Revenue Engines," I recommend prioritizing infrastructure investment and resource allocation here. By strengthening logistics and support services in these established hubs, you can maximize ROI and defend your market share against competitors.

2. Premium Market Segmentation: To capitalize on high-wealth clusters, I recommend launching exclusive "Curated Exclusives" collections. By creating a dedicated, high-end UI section for luxury dining, you provide a clear value proposition for Zomato Gold subscriptions, targeting the specific needs of High-Net-Worth Individuals.

3. Digital Maturity Index: For "Digitally Mature" markets, I recommend shifting marketing budgets toward high-conversion performance-based digital ads. Conversely, in low-maturity regions, the focus should shift toward consumer education and aggressive vendor-onboarding campaigns to build the initial infrastructure.

4. High-Conversion "Value Gems": I recommend creating a dedicated "Budget-Friendly Favorites" collection on the app's homepage. By highlighting these high-quality, low-cost restaurants, you create a "hook" that lowers the barrier for new users, driving acquisition and frequent repeat orders.

5. Market Expansion Opportunities: To capture first-mover advantage, I recommend mobilizing your Sales and Vendor Acquisition teams to execute targeted onboarding campaigns in these low-density, high-activity regions. This effectively turns potential demand into supply, closing the gap in your operational network.

6. Brand Advocacy: I recommend formalizing a "Zomato Trusted Partner" program. By identifying restaurants with exceptional vote-to-rating ratios, you can leverage them as brand influencers—providing them priority listing or exclusive access to new app features in exchange for promoting the platform.

7. Quality & Reliability Benchmarking: I recommend utilizing these "Gold Standard" performers in your B2B training programs. Showcasing these top-tier restaurants to underperforming vendors acts as a "Success Roadmap," demonstrating exactly how high service standards correlate with increased order volumes.

8. Menu Optimization & Demand Forecasting: I recommend implementing a "Smart Menu Consulting" feature for vendors. By providing new partners with automated, data-driven insights on which cuisine combinations are currently trending, you reduce the "failure-to-launch" risk for new businesses and ensure menu alignment with local demand.

9. New Vendor Support ("Cold Start"): To solve the "Cold Start" problem, I recommend launching a "New Venue Spotlight" program. Automatically highlighting new restaurants on the homepage for their first 14 days, combined with a "Try New" discount, will accelerate the collection of initial reviews and boost early platform engagement.

10. Historical Market Velocity: I recommend using historical growth trends for predictive resource planning. By correlating past growth spurts with economic cycles, your strategy team can proactively anticipate hiring needs for delivery fleets and support staff before the next predicted expansion phase begins.